

Edexcel International AS Chemistry



Your notes

Physical Chemistry Core Practicals

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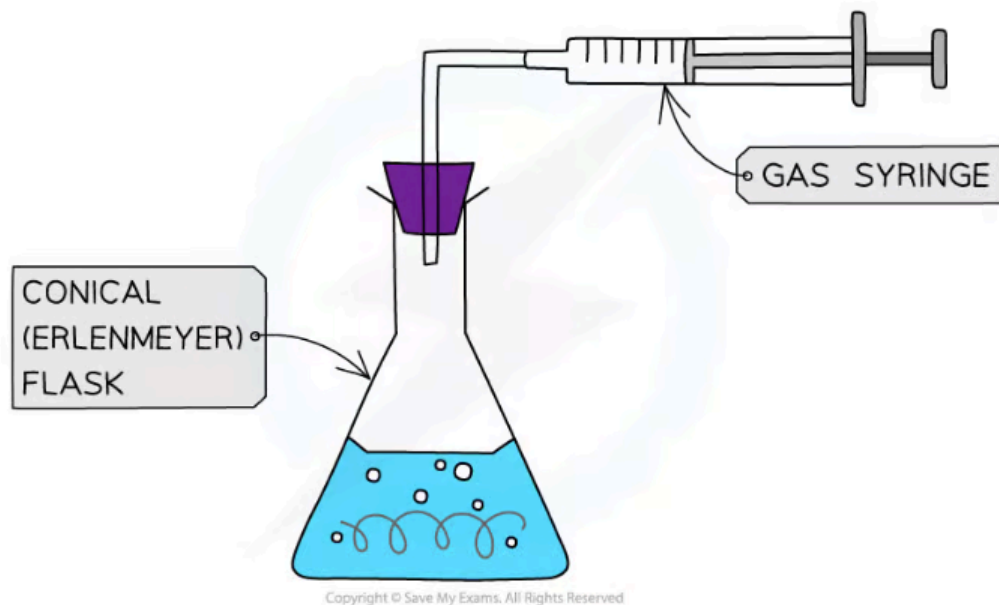
- * Molar Volume of a Gas
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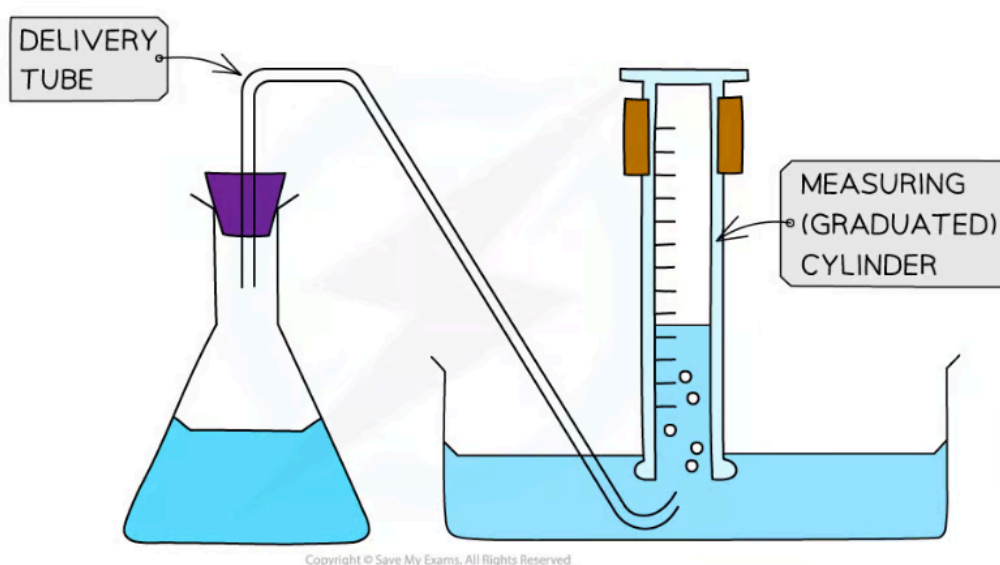
Core Practical 1: Measuring the Molar Volume of a Gas

Measuring gas volumes

- The volume of gas produced in a reaction can be measured by collecting the gas with a gas syringe or by the displacement of water



Gas syringe equipment for collecting the gas produced in a reaction



Displacement of water equipment for collecting the gas produced in a reaction



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Sample method

- For the reaction of hydrochloric acid and sodium carbonate



- Measure out a fixed volume of hydrochloric acid, e.g. 25.0 cm^3 , into a conical flask
- Add a known mass of sodium carbonate, e.g. 0.05 g , to the conical flask
- Immediately connect the gas syringe delivery tube
- Allow the reaction to go to completion
- Record the volume of carbon dioxide produced
- Repeat the experiment with different masses of sodium carbonate, e.g. 0.10 g , 0.15 g , 0.20 g , 0.25 g ... 0.50 g
- Some assumptions are made about the experiment:
 - The amount of gas lost between adding the sodium carbonate and connecting the delivery tube is negligible
 - The delivery tube set up is airtight so no gas is lost
 - The reaction does go to completion

Sample results

Mass Volume Results Table

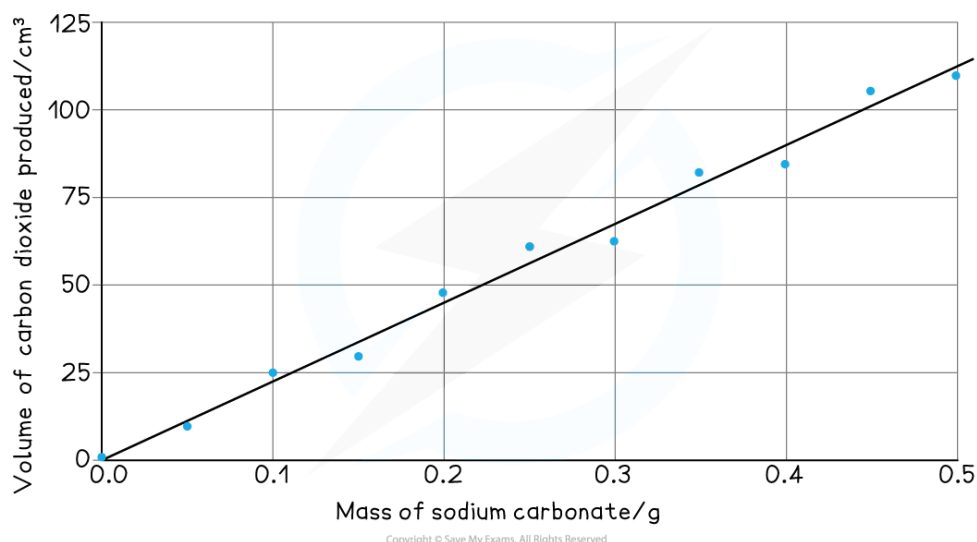
Mass of sodium carbonate/g	Volume of carbon dioxide produced/ cm^3
0.05	9.6
0.10	24.1
0.15	29.2
0.20	47.3
0.25	60.7
0.30	62.5
0.35	79.0
0.40	84.6
0.45	105.4
0.50	110.1

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- The results are then plotted on to a graph
 - Mass of sodium carbonate on the x-axis and volume of carbon dioxide produced on the y-axis
- Anomalous results are ignored and one straight line (or one smooth curve) of best fit is added



Analysis

- Read off the volume of gas produced for a sensible mass of sodium carbonate, e.g. 0.35 g produces 79.0 cm³
 - The mass of sodium carbonate may be specified in an exam question



- From the reaction equation, one mole of sodium carbonate produces one mole of carbon dioxide
- Calculate the molar mass of sodium carbonate
 - $(2 \times 23.0) + 12.0 + (3 \times 16.0) = 106.0$
- Calculate the number of moles of sodium carbonate, using the mass from your graph reading

$$\text{Moles} = \frac{\text{mass}}{\text{molar mass}} = \frac{0.35}{106.0} = 0.0033 \text{ moles}$$

- Convert the volume of carbon dioxide from your graph reading from cm³ to dm³

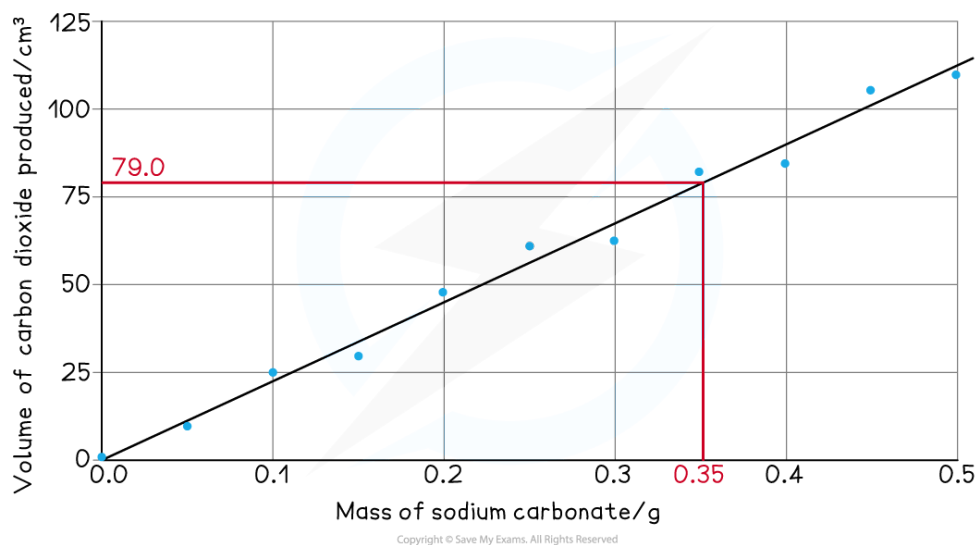
$$\frac{79.0 \text{ cm}^3}{1000} = 0.079 \text{ dm}^3$$

- Calculate the molar volume of gas produced:

$$\blacksquare \text{ Molar gas volume} = \frac{\text{volume}}{\text{moles}} = \frac{0.079}{0.0033} = 23.93 \text{ dm}^3$$



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Application

- This experiment can be used to determine the identity of an unknown metal, M, in a metal carbonate, MCO₃
- This process can be applied to thermal decomposition of metal carbonates as well as their reaction with acid



Worked Example

At room temperature and pressure, 0.950 g of a Group 2 metal carbonate, MCO₃, reacted with hydrochloric acid to produce 226.0 cm³ of carbon dioxide.

Deduce the identity of the metal **M**.

Answer:

Step 1: Find the number of moles of carbon dioxide released using the volume produced at room temperature and pressure:

$$\blacksquare \text{ number of moles of CO}_2 = \frac{\text{volume of gas (dm}^3\text{)}}{\text{molar gas volume (dm}^3\text{)}}$$

$$\blacksquare n(\text{CO}_2) = \frac{0.226}{24} = 0.009417 \text{ mol}$$

Step 2: Find the number of moles of metal carbonate, MCO₃

- One mole of metal carbonate will release one mole of carbon dioxide
 - Number of moles of CO₂ = number of moles of MCO₃
 - n(MCO₃) = 0.009417 mol



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Step 3: Calculate the molar mass of MCO_3

$$M_r = \frac{\text{mass}}{\text{moles}}$$

$$M_r = \frac{0.950}{0.009417} = 100.9 \text{ g mol}^{-1}$$

Step 4: Calculate the atomic mass of M in MCO_3 and deduction of the Group 2 metal

- $M_r = \Sigma(\text{atomic masses})$
 - $100.9 = M + 12.0 + (3 \times 16.0)$
 - $M = 100.9 - 60.0 = 40.9 \text{ g mol}^{-1}$
 - The closest Group 2 atomic mass is calcium at 40.1 g mol^{-1} , therefore the metal M is calcium



Examiner Tips and Tricks

Careful: Examiners can write these questions to include the following distractions:

- The molar mass of the metal carbonate / MCO_3 is close to the mass of a Group 2 metal
 - The mass of the carbonate ion needs to be subtracted from the molar mass in order to deduce the identity of the metal
- The atomic mass of the metal is close the atomic mass of another metal, not necessarily a Group 2 metal
 - Read the question as it will provide information about the metal

The above points can be applied to any metal carbonate, not just Group 2 metal carbonates although they are the most common

Hazards, risks and precautions



**HARMFUL TO
HEALTH**



CORROSIVE



OXIDISING

- The hazards associated with acids depend on the type and concentration of the acid



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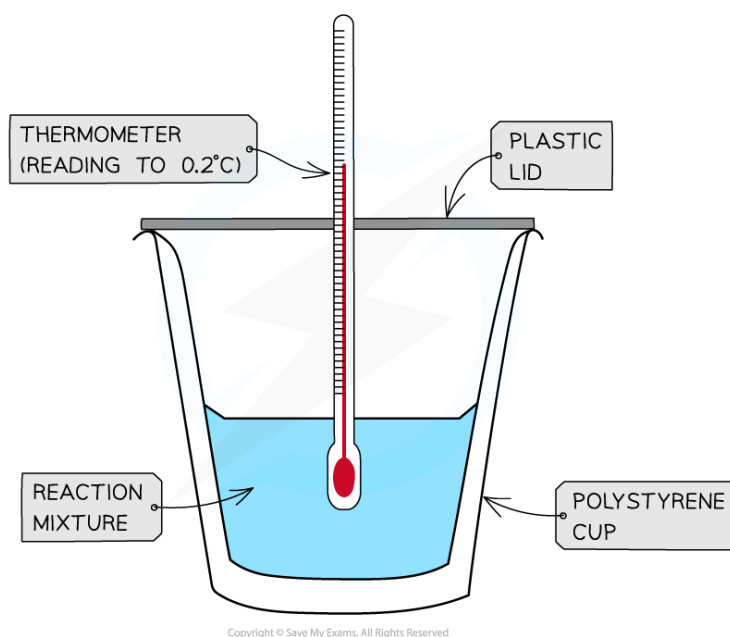
- Most dilute acids either require no hazard symbol or they are an irritant, so require the symbol to show they are harmful to health
 - Eye protection should be worn when handling
- Moderately concentrated acids are often corrosive
 - In addition to eye protection, gloves should also be worn
- Some concentrated acids, e.g. nitric acid, are oxidising which can cause or intensify a fire in contact with combustible materials
 - Eye protection and gloves are necessary when handling concentrated acids and the use of a fume cupboard is often required



Core Practical 2: Determining Enthalpy Change of Reaction

Measuring enthalpy changes

- **Calorimetry** is a technique used to measure changes in enthalpy of chemical reactions
- A **calorimeter** can be made up of a **polystyrene drinking cup**, a **vacuum flask** or **metal can**



A polystyrene cup can act as a calorimeter to find enthalpy changes in a chemical reaction

The energy needed to raise the temperature of 1 g of a substance by 1 K is called the **specific heat capacity** (c) of the liquid. The **specific heat capacity** of water is $4.18 \text{ J g}^{-1} \text{ K}^{-1}$. The energy transferred as heat can be calculated by:

$$q = m \times c \times \Delta T$$

q = THE HEAT TRANSFERRED, J
 m = THE MASS OF WATER, g
 c = THE SPECIFIC HEAT CAPACITY, $\text{J g}^{-1} \text{ K}^{-1}$
 ΔT = THE TEMPERATURE CHANGE, K

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Equation for calculating energy transferred in a calorimeter

Enthalpy changes for reactions in solution



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- The principle of these calorimetry experiments is to carry out the reaction with an excess of one reagent and measure the temperature change over the course of a few minutes
- The apparatus needed to carry out an enthalpy of reaction in solution calorimetry experiment is shown above

Sample method for a displacement reaction

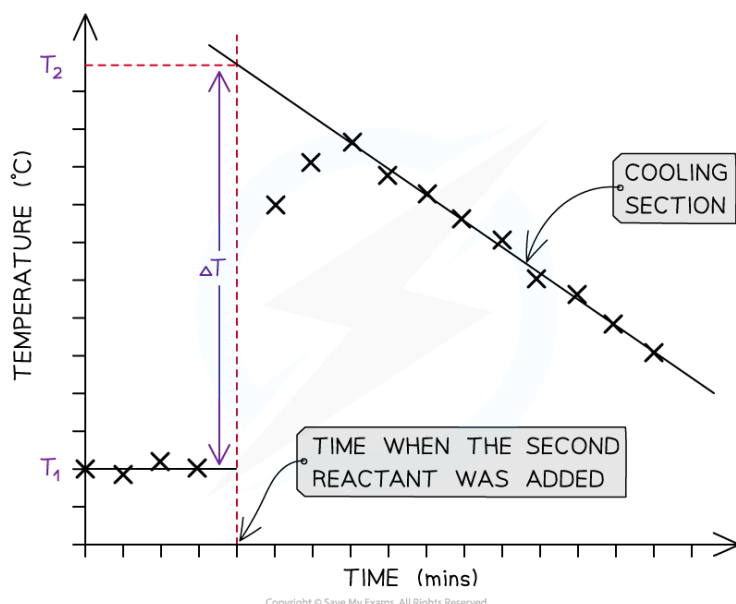
1. Using a measuring cylinder place 25 cm^3 of the 1.0 mol dm^{-3} copper(II) sulphate solution into the polystyrene cup
 2. Weigh about 6 g of zinc powder – as this is an excess there is no need to be very accurate
 3. Draw a table to record the initial temperature and then the temperature and time every half minute up to 9.5 minutes
 4. Put a thermometer or temperature probe in the cup, stir, and record the temperature every half minute for 2.5 minutes
 5. At precisely 3 minutes, add the zinc powder to the cup (DO NOT RECORD THE TEMPERATURE AT 3 MINUTES)
 6. Continue stirring and record the temperature for an additional 6 minutes
- For the purposes of the calculations, some assumptions are made about the experiment:
 - That the specific heat capacity of the solution is the same as pure water, i.e. $4.18\text{ J g}^{-1}\text{ K}^{-1}$
 - That the density of the solution is the same as pure water, i.e. 1 g cm^{-3}
 - The specific heat capacity of the container is ignored
 - The reaction is complete
 - There are negligible heat losses

Temperature correction graphs

- For reactions which are not instantaneous there may be a delay before the maximum temperature is reached
- During that delay the substances themselves may be losing heat to the surroundings, so that the true maximum temperature is never actually reached
- To overcome this problem we can use graphical analysis to determine the maximum enthalpy change



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A temperature correction graph for a metal displacement reaction between zinc and copper sulfate solution. The zinc is added after 4 minutes

The steps to make a temperature correction graph are:

1. Take a temperature reading before adding the reactants for a few minutes to get a steady value
2. Add the second reactant and continue recording the temperature and time
3. Plot the graph and **extrapolate** the cooling part of the graph until you intersect the time at which the second reactant was added

Analysis

- Use both extrapolated lines to calculate ΔT as shown on the graph
- Use the equation $q = mc\Delta T$ to calculate the energy transferred
 - q = energy transferred
 - m = mass - this will be the mass of the 25 cm^3 solution which will be 25 g (assuming a density of 1 g cm^{-3})
 - c = specific heat capacity - this will be assumed to be $4.18 \text{ J g}^{-1} \text{ K}^{-1}$, which is the specific heat capacity of water
 - ΔT = the temperature change from the graph
- Convert your value for energy transferred from J into kJ
- Then use the equation $\Delta H = \frac{q}{n}$ to calculate the enthalpy change for the reaction
 - q = energy transferred

- n = number of moles - this would be the number of moles of the **limiting reagent**, which means that you will have an extra calculation to do to determine whether this is the zinc or the copper sulfate
- Remember that in the example above, the temperature of the reaction mixture increased which means that the reaction is exothermic and should, therefore, have a negative value



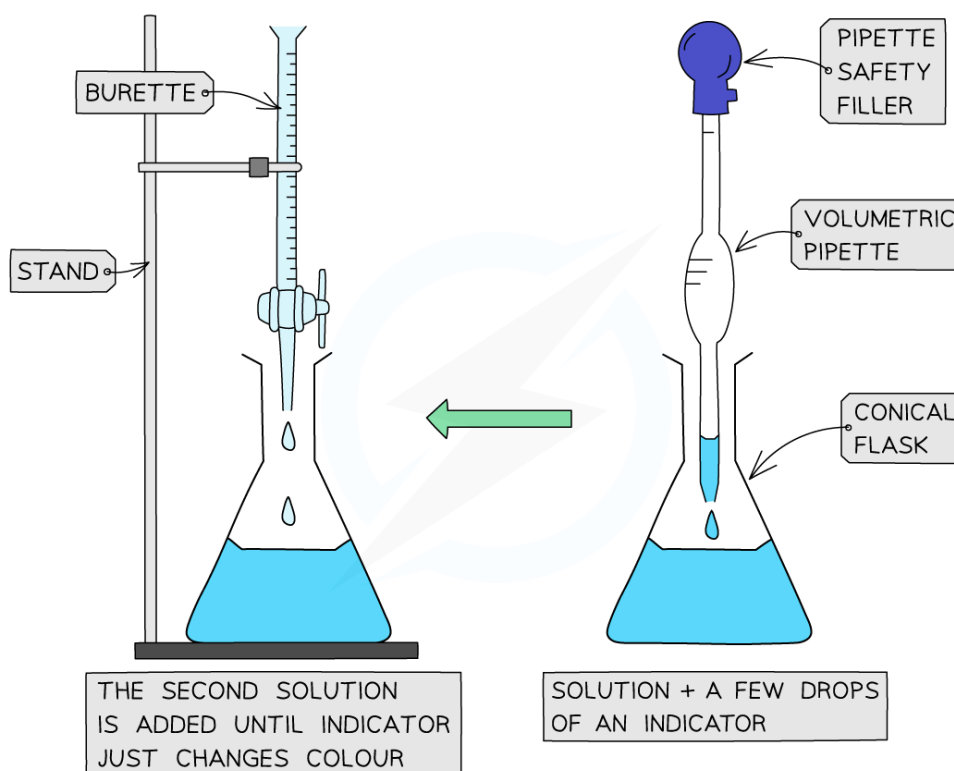
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Core Practical 3: Hydrochloric Acid Concentration

Performing the Titration

- The key piece of equipment used in the titration is the burette
- **Burettes** are usually marked to a precision of 0.10 cm^3
 - Since they are analogue instruments, the uncertainty is recorded to **half** the smallest marking, in other words to $\pm 0.05 \text{ cm}^3$
- The **end point** or **equivalence point** occurs when the two solutions have reacted completely and is shown with the use of an **indicator**



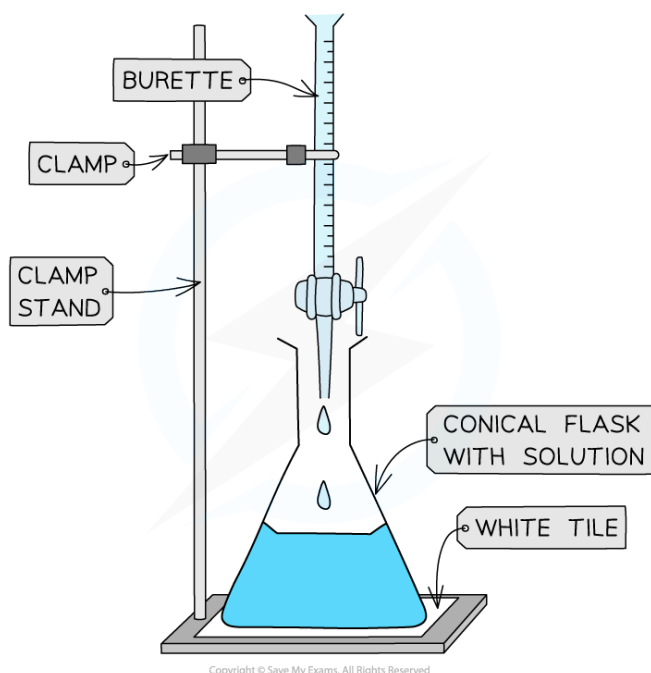
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The steps in a titration

- A white tile is placed under the conical flask while the titration is performed, to make it easier to see the colour change



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Titrating

- The steps in a titration are:
 - Measuring a known volume (usually 20 or 25 cm³) of one of the solutions with a **volumetric pipette** and placing it into a **conical flask**
 - The other solution is placed in the **burette**
 - To start with, the burette will usually be filled to 0.00 cm³
 - A few drops of the **indicator** are added to the solution in the conical flask
 - The tap on the **burette** is carefully opened and the solution added, portion by portion, to the **conical flask** until the **indicator** starts to change colour
 - As you start getting near to the end point, the flow of the burette should be slowed right down so that the solution is added dropwise
 - You should be able to close the tap on the burette after one drop has caused the colour change
 - Multiple runs are carried out until **concordant** results are obtained
 - Concordant results are within 0.1 cm³ of each other

Recording and processing titration results

- Both the initial and final **burette** readings should be recorded and shown to a **precision** of $\pm 0.05 \text{ cm}^3$, the same as the **uncertainty**



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ALL RESULTS ARE RECORDED TO 2 DECIMAL PLACES INCLUDING ZERO READINGS

	Rough	Run 1	Run 2	Run 3
Initial burette reading ± 0.05 ml	0.00	23.15	0.20	23.00
Final burette reading ± 0.05 ml	23.75	45.95	23.00	46.10
Volume delivered ± 0.10 ml	23.75	22.80 ✓	22.80 ✓	23.10

THE FINAL DIGIT IS 0 OR 5

DOUBLE THE UNCERTAINTY

THE ROUGH RESULT IS USUALLY FAR OVER THE END-POINT

THIS RESULT IS DISCARDED AS IT IS TOO HIGH

✓ = CONCORDANT RESULTS

USED TO CALCULATE THE AVERAGE

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A typical layout and set of titration results

- The volume delivered (**titre**) is calculated and recorded to an **uncertainty** of $\pm 0.10 \text{ cm}^3$
 - The **uncertainty** is doubled, because two **burette** readings are made to obtain the **titre** ($V_{\text{final}} - V_{\text{initial}}$), following the rules for **propagation of uncertainties**
- Concordant** results are then averaged, and non-concordant results are discarded
- The appropriate calculations are then done



Worked Example

25.0 cm^3 of hydrochloric acid was titrated with a $0.200 \text{ mol dm}^{-3}$ solution of sodium hydrogencarbonate, NaHCO_3 .



Use the following results to calculate the concentration of the acid, to 3 significant figures.



Your notes

	Rough	Run 1	Run 2	Run 3
Initial burette reading ± 0.05 ml	0.00	23.15	0.20	23.00
Final burette reading ± 0.05 ml	23.75	45.95	23.00	46.10
Volume delivered ± 0.10 ml	23.75	22.80	22.80	23.10

Answer

Step 1: Calculate the average titre

$$\text{Average titre} = \frac{22.80 + 22.80}{2} = 22.80 \text{ cm}^3$$

Step 2: Calculate the number of moles of sodium hydrogencarbonate

$$\text{Moles} = \frac{22.80}{1000} \times 0.200 = 4.56 \times 10^{-3} \text{ moles}$$

Step 3: Calculate (or deduce) the number of moles of hydrochloric acid

- The stoichiometry of $\text{NaHCO}_3 : \text{HCl}$ is 1 : 1
 - Therefore, the number of moles of sodium hydrogencarbonate is also 4.56×10^{-3} moles

Step 4: Calculate the concentration of hydrochloric acid

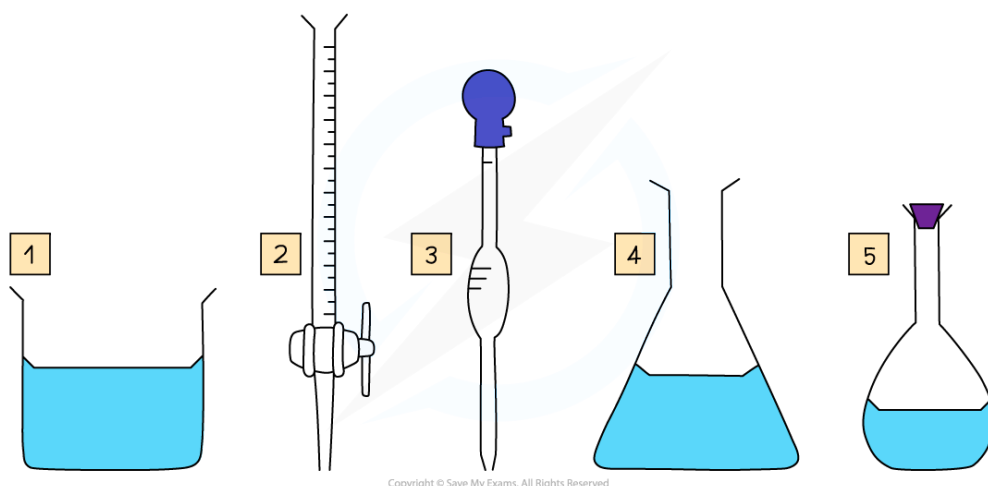
$$\text{Concentration} = \frac{\text{moles}}{\text{volume}} = \frac{4.56 \times 10^{-3}}{(25.0 / 1000)} = 0.182 \text{ mol dm}^{-3}$$



Core Practical 4: Preparing a Standard Solution

Volumetric Analysis

- **Volumetric analysis** is a process that uses the volume and concentration of one chemical reactant (**a volumetric solution**) to determine the concentration of another unknown solution
- The technique most commonly used is a **titration**
- The volumes are measured using two precise pieces of equipment, a **volumetric** or **graduated pipette** and a **burette**
- Before the titration can be done, the standard solution must be prepared
- Specific apparatus must be used both when preparing the standard solution and when completing the titration, to ensure that volumes are measured precisely



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Some key pieces of apparatus used to prepare a volumetric solution and perform a simple titration

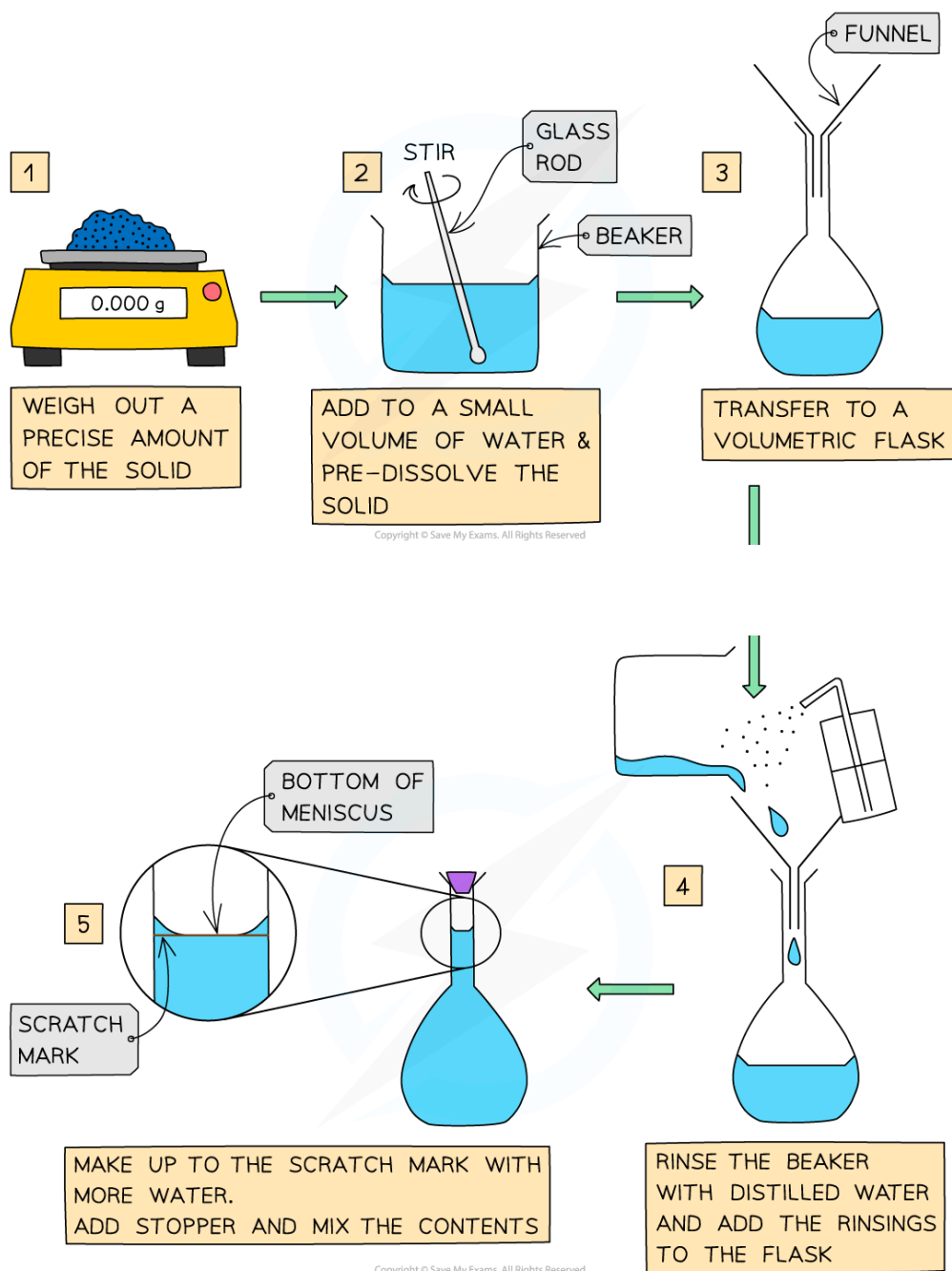
1. Beaker
2. Burette
3. Volumetric Pipette
4. Conical Flask
5. Volumetric Flask

Making a Volumetric Solution

- Chemists routinely prepare solutions needed for analysis, whose concentrations are known precisely
- These solutions are termed **volumetric solutions** or **standard solutions**
- They are made as accurately and precisely as possible using three decimal place balances and volumetric flasks to reduce the impact of measurement uncertainties
- The steps are:



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Volumes & concentrations of solutions



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- The **concentration** of a solution is the amount of **solute** dissolved in a **solvent** to make 1 dm^3 of **solution**
 - The solute is the substance that dissolves in a solvent to form a solution
 - The solvent is often water
- A **concentrated** solution is a solution that has a **high** concentration of solute
- A **dilute** solution is a solution with a **low** concentration of solute
- Concentration is usually expressed in one of three ways:
 - moles per unit volume
 - mass per unit volume
 - parts per million



Worked Example

Calculate the mass of sodium hydroxide, NaOH, required to prepare 250 cm^3 of a $0.200 \text{ mol dm}^{-3}$ solution

Answer:

Step 1: Find the number of moles of NaOH needed from the concentration and volume:

- $\text{number of moles} = \text{concentration (mol dm}^{-3}) \times \text{volume (dm}^3)$
 - $n = 0.200 \text{ mol dm}^{-3} \times 0.250 \text{ dm}^3$
 - $n = \mathbf{0.0500 \text{ mol}}$

Step 2: Find the molar mass of NaOH

- $M = 22.99 + 16.00 + 1.01 = 40.00 \text{ g mol}^{-1}$

Step 3: Calculate the mass required

- $\text{mass} = \text{moles} \times \text{molar mass}$
 - $\text{mass} = 0.0500 \text{ mol} \times 40.00 \text{ g mol}^{-1} = \mathbf{2.00 \text{ g}}$